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B.M.S. College of Engineering, Bengaluru-560019

Autonomous Institute Affiliated to VTU

May / June 2019 Semester End Main Examinations

Programme: B.E. **Semester : IV**
Branch : Information Science and Engineering **Duration: 3 hrs.**
Course Code: 15IS4DCADA **Max Marks: 100**
Course: Analysis and Design of Algorithms **Date: 30.05.2019**

Instructions: 1. Answer any FIVE full questions, choosing one full question from each unit.
2. Missing data, if any, may suitably assumed.

Important Note: Completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages. Revealing of identification, appeal to evaluator will be treated as malpractice.

UNIT - I

1 a) Explain the three asymptotic methods used to represent the rate of growth of running time of an algorithm. Show that $2n^2+3n+1 \in \theta(n^2)$ **08**

b) What does the following algorithm compute? What is its efficiency class? **06**

Calculate(n)
S ← 0
for i ←1 to n do
*S ← S + i*i*
return S

c) If $f_1(t) \in O(g_1(t))$ and $f_2(t) \in O(g_2(t))$, prove that $f_1(t)+f_2(t) \in O(\max(g_1(t),g_2(t)))$. **06**

UNIT - II

2 a) Two words are anagrams of one another if their letters can be rearranged to form the other word. For example, 'tea' and 'eat' are anagrams. Design a brute force based algorithm to verify if two given string are anagrams or not. **06**

b) Explain with pseudo code of how quicksort works to sort an array. **08**

c) Apply divide and conquer design strategy to multiply two integers: 5327 and 7964 **06**

OR

3 a) State master's theorem and use the same to get tight asymptotic bounds for the recurrence relation: $T(n) = 4\ T(n/2) + n^3$ **06**

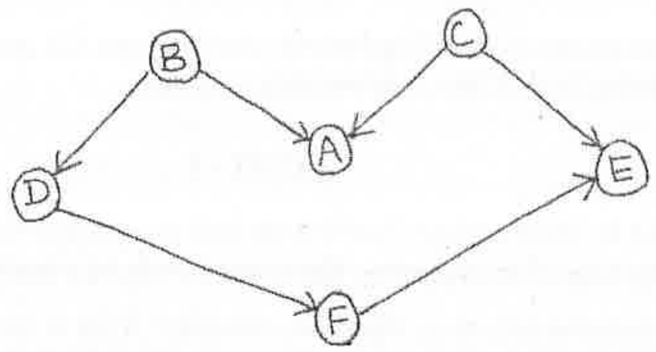
b) Trace quicksort for the following dataset: 65,70,75,80,85,60,55,50,45. Also draw the tree for the recursive calls made. **08**

c) Apply Strassen's algorithm to compute – 06

$$\begin{vmatrix} 5 & 2 \\ 3 & 5 \end{vmatrix} \times \begin{vmatrix} 7 & 1 \\ 9 & 0 \end{vmatrix}$$

UNIT - III

4 a) Solve topological sorting problem for the given graph using DFS based algorithm. 06



b) Design a pre-sorting based algorithm to find the smallest and largest elements in an array of n numbers. Determine it efficiency class. Is it better than the brute force algorithm? 07

c) Construct an AVL tree by inserting the elements 5,4,2,3,7,6,1. 07

OR

5 a) Write the algorithm for Insertion sort. Comment on its best and worst case running time. 07

b) Differentiate between DFS and BFS algorithms. 05

c) Sort the list 3,5,6,2,4,7,1 using heapsort. Show the heapification at every step. 08

UNIT - IV

6 a) Apply Horspool algorithm to search for the pattern BARBER in the text JIM_SAW_ME_IN_THE_BARBER_SHOP 08

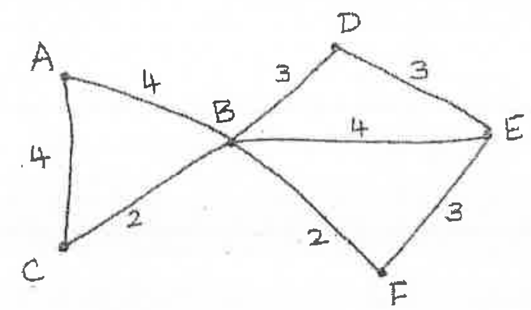
b) Solve the following instance of Knapsack problem using dynamic programming given that the capacity of the knapsack is 5. 07

Item	Weight	Value
1	2	12
2	1	10
3	3	20
4	2	15

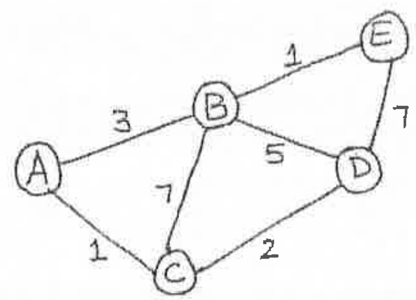
c) With the pseudo code for Floyd's algorithm, explain of how it would solve the all-pairs shortest path problem. 05

UNIT - V

7 a) Apply Kruskal's algorithm for the following graph to identify the Minimum Spanning Tree. 07



b) Solve the following instance of single-source shortest path problem with vertex C as the source. 08



c) Define P class and NP class of problems with an example for each. 05

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